

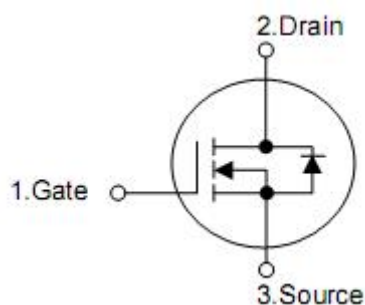
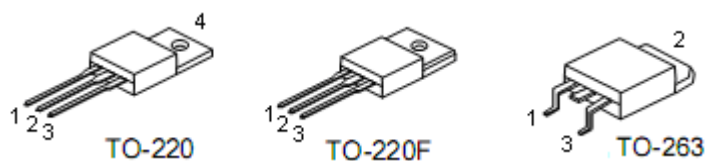
1. Description

The power MOSFET is produced using KIA semi's advanced planar stripe DMOS technology. This advanced technology has been especially tailored to minimize on-state resistance, provide superior switching performance, and withstand high energy pulse in the avalanche and commutation mode. These devices are well suited for high efficiency switched mode power supplies, active power factor correction based on half bridge topology.

2. Features

- n** $R_{DS(on)}=1.2\Omega$ @ $V_{GS}=10V$
- n** Low gate charge (typical 29nC)
- n** High ruggedness
- n** Fast switching
- n** 100% avalanche tested
- n** Improved dv/dt capability

3. Pin configuration



Pin	Function
1	Gate
2	Drain
3	Source

4. Absolute maximum ratings

(T_C=25°C , unless otherwise noted)

Parameter		Symbol	Rating			Units
			TO263	TO220	TO220F	
Drain-source voltage		V _{DSS}	650			V
Gate-source voltage		V _{GSS}	±30			V
Drain current continuous	T _C =25°C	I _D	7.0	7.0	7.0*	A
	T _C =100°C		4.2	4.2	4.2*	A
Drain current pulsed (note1)		I _{DM}	28	28	28*	A
Avalanche energy	Repetitive (note1)	E _{AR}	14.7			mJ
	Single Pulse (note2)	E _{AS}	230			mJ
Peak diode recovery dv/dt (note3)		dv/dt	4.5			V/ns
Total power dissipation	T _C =25°C	P _D	147	147	48	W
	Derate above 25°C		1.18	1.18	0.38	W/°C
Operating and storage temperature range		T _J ,T _{STG}	-55~+150			°C
Maximum lead temperature for soldering purposes, 1/8"from case for 5 seconds		T _L	300			°C

5. Thermal characteristics

Parameter	Symbol	Rating			Unit
		TO263	TO220	TO220F	
Thermal resistance junction-ambient	R _{thJA}	62.5			°C/W
Thermal resistance,case-to-sink typ.	R _{thCS}	0.5	0.5	-	
Thermal resistance junction-case	R _{thJC}	0.85	0.85	2.6	

6. Electrical characteristics

(T_J=25°C, unless otherwise noted)

Parameter		Symbol	Test Conditions	Min	Typ	Max	Units
Off characteristics							
Drain-source breakdown voltage		BV _{DSS}	V _{GS} =0V,I _D =250μA	650	-	-	V
Zero gate voltage drain current		I _{DSS}	V _{DS} =650V,V _{GS} =0V	-	-	1	μA
			V _{DS} =520V,T _C =125°C	-	-	10	μA
Gate-body leakage current	Forward	I _{GSS}	V _{GS} =30V,V _{DS} =0V	-	-	100	nA
	Reverse		V _{GS} =-30V,V _{DS} =0V	-	-	-100	nA
Breakdow voltage temperature coefficient		ΔBV _{DSS} ΔT _J	I _D =250μA, Referenced to25 °C	-	0.7	-	V/ °C
On characteristics							
Gate threshold voltage		V _{GS(TH)}	V _{DS} =V _{GS} ,I _D =250μA	2.0	-	4.0	V
Static drain-source on-resistance		R _{DS(ON)}	V _{DS} =10V,I _D =3.5A	-	1.2	1.4	Ω
Dynamic characteristics							
Input capacitance		C _{ISS}	V _{DS} =25V,V _{GS} =0V, f=1MHz	-	1000	-	pF
Output capacitance		C _{OSS}		-	110	-	pF
Reverse transfer capacitance		C _{RSS}		-	12.6	-	pF
Switching characteristics							
Turn-on delay time		t _{D(ON)}	V _{DD} =325V,R _G =25Ω, I _D =7.0A (note4,5)	-	20	-	ns
Rise time		t _R		-	50	-	ns
Turn-off delay time		t _{D(OFF)}		-	80	-	ns
Fall time		t _F		-	70	-	ns
Total gate charge		Q _G	V _{DS} =520V,V _{GS} =10V I _D =7.0A (note4,5)	-	29	-	nC
Gate-source charge		Q _{GS}		-	4.7	-	nC
Gate-drain charge		Q _{GD}		-	12.5	-	nC
Drain-source diode characteristics							
Drain-source diode forward voltage		V _{SD}	V _{GS} =0V,I _S =7.5A	-	-	1.4	V
Continuous drain-source current		I _S		-	-	7.0	A
Pulsed drain-source current		I _{SM}		-	-	28	A
Reverse recovery time		t _{RR}	I _S =7.5A,	-	350	-	ns
Reverse recovery charge		Q _{RR}	dI _{SD} /dt=100A/μs (note 4)	-	3.3	-	μC

Note: 1. Repetitive rating: pulse width limited by maximum junction temperature

2. L=7.3mH, I_{AS}=7.0A, V_{DD}=50V, R_G=25Ω, starting T_J=25°C

3. I_{SD}≤7.0A, di/dt≤200A/μs, V_{DD}≤BV_{DSS}, starting T_J=25°C

4. Pulse test: pulse width ≤300μs, duty cycle ≤2%

5. Essentially independent of operating temperature

7. Test circuits and waveforms

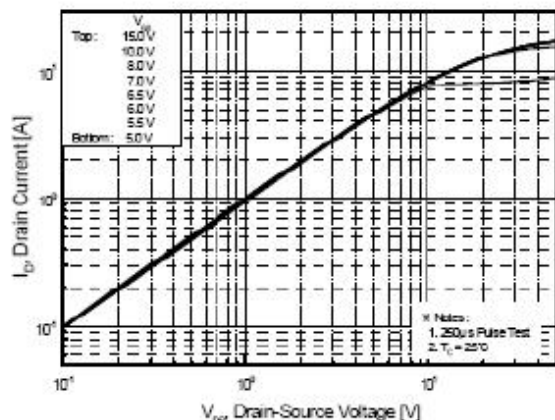


Figure 1. On-Region Characteristics

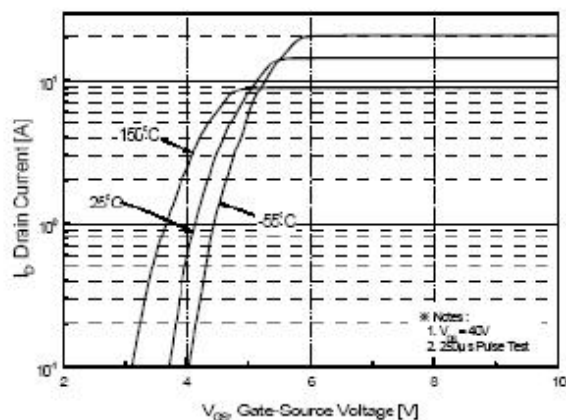


Figure 2. Transfer Characteristics

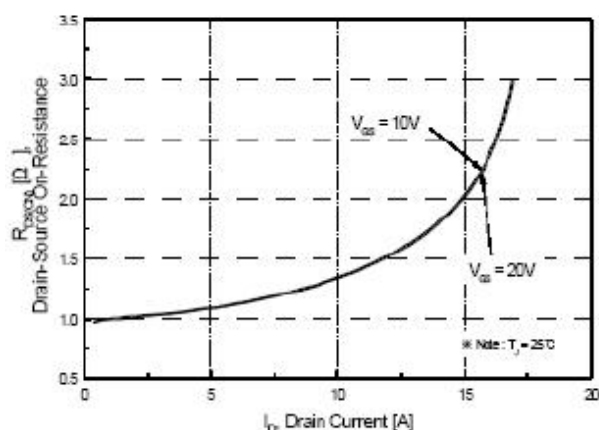


Figure 3. On-Resistance Variation vs Drain Current and Gate Voltage

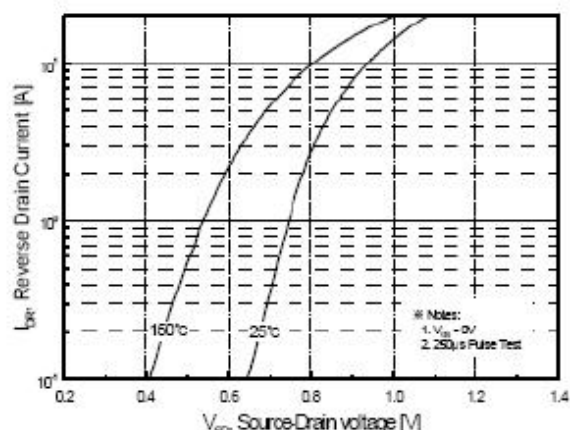


Figure 4. Body Diode Forward Voltage Variation with Source Current and Temperature

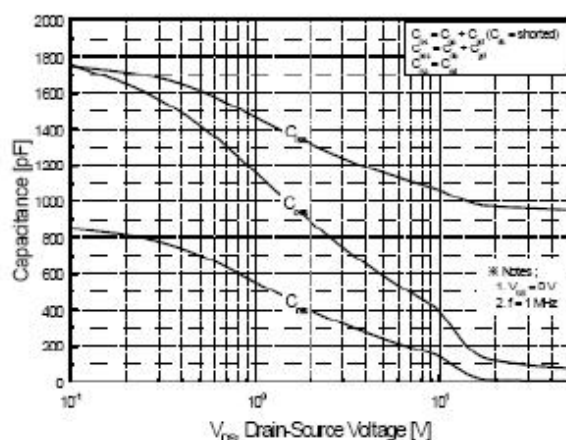


Figure 5. Capacitance Characteristics

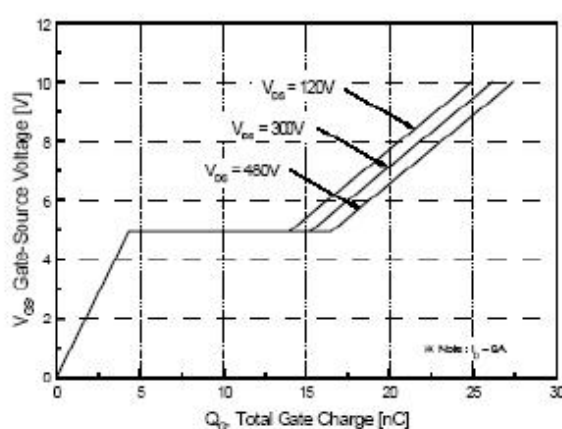


Figure 6. Gate Charge Characteristics

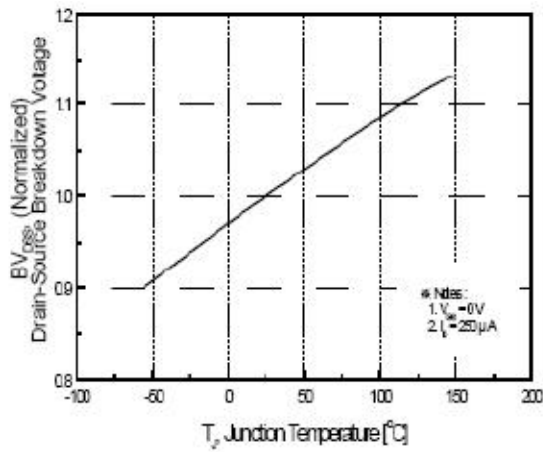


Figure 7. Breakdown Voltage Variation vs Temperature

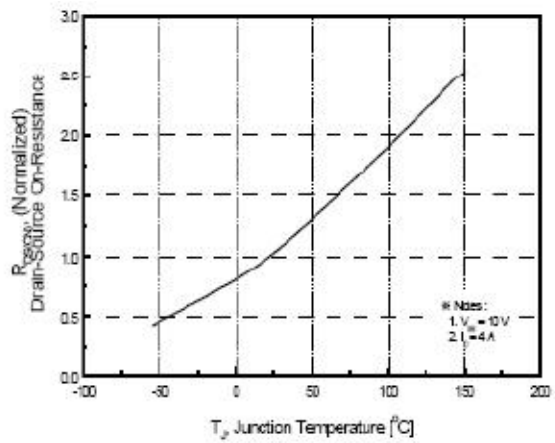


Figure 8. On-Resistance Variation vs Temperature

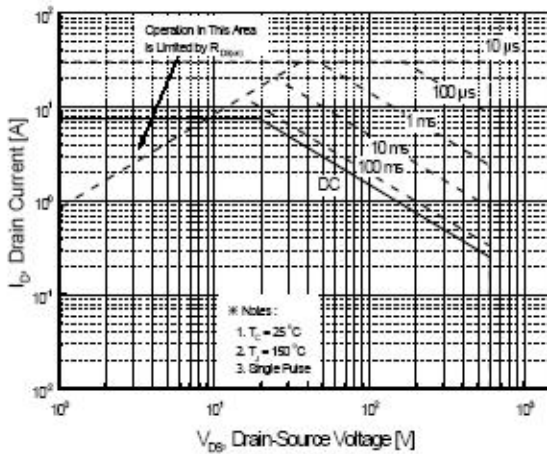


Figure 9-1. Maximum Safe Operating Area for TO-263 TO-220

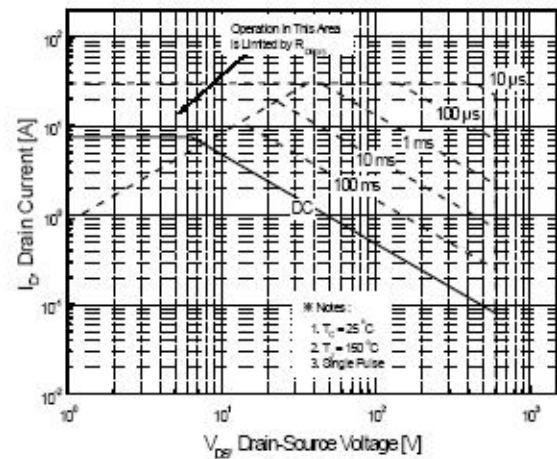


Figure 9-2. Maximum Safe Operating Area for TO-220F

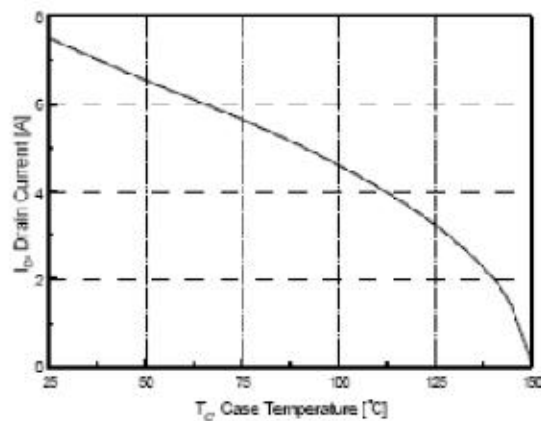
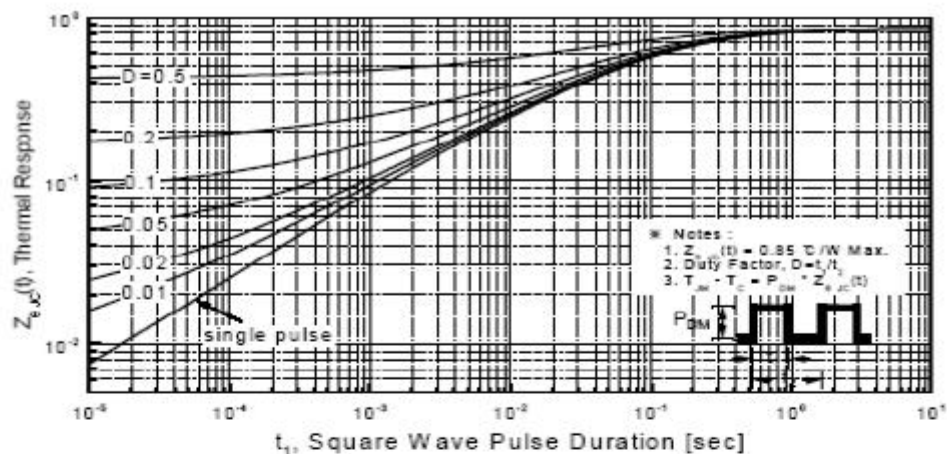
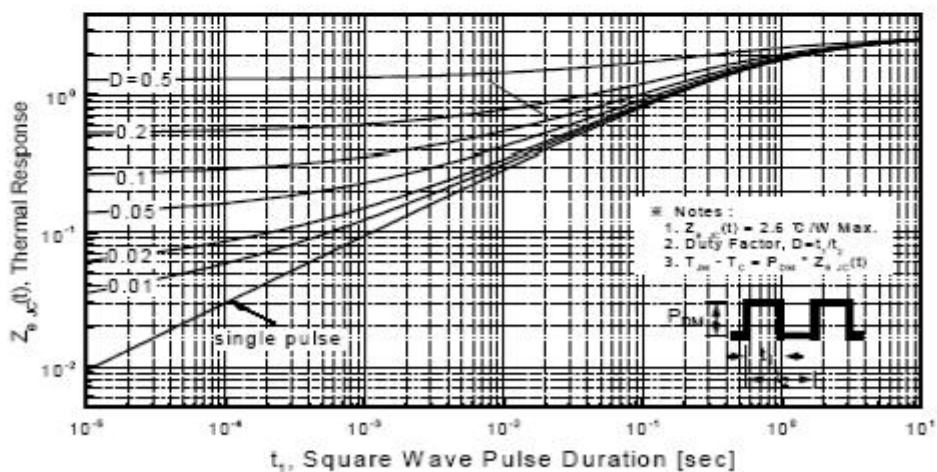


Figure 10. Maximum Drain Current vs Case Temperature



**Figure 11-1. Transient Thermal Response Curve
for TO-263 TO-220**



**Figure 11-2. Transient Thermal Response Curve
for TO-220F**